

Lab Booking

CITS3001 Advanced Algorithms — Time limit: 3s

The department has one bookable lab and a pile of booking requests, each with a start time and an end time. Two bookings are compatible if they do not overlap — a booking that starts exactly when another ends is fine. Accept as many bookings as possible.

Input

The first line contains an integer n ($1 \leq n \leq 2 \cdot 10^5$), the number of requests. Each of the next n lines contains two integers s and e ($0 \leq s < e \leq 10^9$), the start and end time of one request.

Output

Output the maximum number of requests that can be accepted such that no two accepted bookings overlap.

Sample Input 1

```
4
1 3
2 4
3 5
6 7
```

Sample Output 1

```
3
```

Sample Input 2

```
3
1 10
2 3
4 9
```

Sample Output 2

```
2
```